Calculus

HW6

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Sec 1

Homework

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Find

1) $\int_{-1}^{1}\left(6x^5 - \frac{2}{\sqrt[5]{x^3}} + 3\right)dx$

$2x^{-\frac{3}{5}}$

$$= x^6 - \frac{2x^{\frac{2}{5}}}{\frac{2}{5}} + 3x \Big|_{-1}^{1} = x^6 - 5\sqrt[5]{x^2} + 3x \Big|_{-2}^{1}$$

$$= (1)^6 - 5\sqrt[5]{(1)^2} + 3(1) - \left((-1)^6 - 5\sqrt[5]{(-1)^2} + 3(-1)\right)$$

$$= \cancel{1} - \cancel{5} + 3 - \cancel{1} + \cancel{5} + 3$$

$$= 6$$ #

2) $\int_1^4 |x-3|\,dx \rightarrow \begin{matrix} x-3 \geq 0 \\ x \geq 3 \end{matrix} \quad \begin{cases} x-3, & x \geq 3 \\ -(x-3), & x < 3 \end{cases}$

$$\int_3^4 x-3\,dx + \int_1^3 -(x-3)\,dx$$

$$= \frac{x^2}{2} - 3x \Big|_3^4 + \left(-\frac{x^2}{2} + 3x\right)\Big|_1^3$$

$$= \frac{16}{2} - 12 - \left(\frac{9}{2} - 9\right) + \left(\left(-\frac{9}{2} + 9\right) - \left(-\frac{1}{2} + 3\right)\right)$$

$$= \frac{16 - 24 - 9 + 18 - 9 + 18 + 1 - 6}{2}$$

$$= \frac{5}{2}$$

3) $\int \frac{(2x^2+1)}{(2x^3+3x)^2}\,dx$

$$= \int (2x^2+1)(2x^3+3x)^{-2}\,dx$$

$$= \frac{1}{3}\int (2x^3+3x)^{-2}(6x^2+3)\,dx$$

$$= \frac{1}{3}\left(-\frac{1}{2x^3+3x}\right)+C$$

$$= -\frac{1}{6x^3+9x}+C$$